

STAR-Trap macrophages selectively capture IL-6/sIL-6R α complexes to attenuate IL-6 trans-signaling

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Abstract

IL-6 trans-signaling is mediated by recruitment of widely expressed gp130 by the IL-6/sIL-6R α complex, expanding IL-6 responsiveness and promoting chronic inflammation. Selective inhibition therefore represents an important strategy for anti-inflammatory intervention. We targeted the composite gp130-binding surface formed by IL-6 and sIL-6R α . RFdiffusion and ProteinMPNN generated 27 *de novo* candidate proteins, which were fused to the FcR γ intracellular domain and screened through B cell surface display and flow cytometry to identify a lead sequence. We coupled dCas9-Activation-Induced Cytidine Deaminase* Δ -mediated hypermutation with gp130-caspase9 functional selection, linking B cell somatic hypermutation to survival-based selection and enriching variants with complex-selective binding and gp130-competitive activity, termed STAR6. Surface expression of the evolved protein in macrophages extended competitive blockade to cell-mediated complex capture and clearance. This study harnesses B cell somatic hypermutation for functional evolution of *de novo*-designed proteins and translates selective recognition of a cytokine-receptor complex into cellular clearance, providing a strategy for selectively modulating IL-6 trans-signaling.

Keywords: IL-6 trans-signaling, *de novo* design, somatic hypermutation, AID, cytokine complex clearance

Introduction

IL-6 signaling occurs through two principal modes. In classical signaling, IL-6 first binds membrane-bound IL-6 receptor α (mIL-6R α), which is expressed mainly by hepatocytes and selected leukocyte subsets. By contrast, trans-signaling is initiated when IL-6 forms a complex with soluble IL-6R α (sIL-6R α), which subsequently recruits the ubiquitously expressed gp130. Because gp130 is broadly expressed across virtually all cell types, albeit at variable levels, trans-signaling extends IL-6 responsiveness to cells lacking mIL-6R α , thereby contributing to inflammatory-cell recruitment, vascular endothelial activation, fibrosis, and the maintenance and amplification of chronic inflammation^{1,2}.

Selective targeting of IL-6 trans-signaling may enable the inflammatory amplification loop to be uncoupled from basal immune homeostasis. The key is to obtain binders that selectively recognize the IL-6/sIL-6R α complex over its free components and bind it tightly enough to compete with gp130. Advances in *de novo* protein design have made it possible to design binding proteins against predefined three-dimensional surfaces. However, candidates obtained through *de novo* design generally represent only the starting point for functional optimization^{3,4}. After initial scaffolds have been generated computationally, an evolutionary system capable of continuously generating sequence diversity and subjecting variants to functional selection in a mammalian cellular environment is still required to bridge the gap between structural feasibility and biological function⁵.

Somatic hypermutation (SHM) in B cells and the antibody affinity maturation that it drives constitute a canonical process of molecular evolution in adaptive immunity. During the germinal center reaction, activation-induced cytidine deaminase (AID) continuously introduces high-frequency mutations into immunoglobulin variable regions, generating B cell clones with diverse antigen-binding capacities. High-affinity variants are subsequently enriched through iterative rounds of antigen-dependent selection and clonal expansion⁶. Redirecting this canonical mechanism of B cell antibody evolution towards designed sequences may therefore enable further optimization of the affinity and function of *de novo*-designed proteins.

Based on this rationale, we developed an approach integrating *de novo* computational

design with cell-based directed evolution and established a protein-directed evolution system targeting the IL-6/sIL-6R α complex⁷. Rather than broadly blocking IL-6, this study targeted the gp130-binding interface exposed upon formation of the IL-6/sIL-6R α complex, with the aim of selectively inhibiting pathogenic trans-signaling while preserving the functions of classical IL-6 signaling. We subsequently constructed a dCas9-AID* Δ -mediated targeted hypermutation system to harness the natural SHM machinery of B cells for the continuous evolution of the designed protein. In parallel, we established a functional selection system coupling the gp130 extracellular domain to a caspase9 death-effector module. This system combined fluorescence-based positive selection for high-affinity binding to the IL-6/sIL-6R α complex with cell death-mediated negative selection to exclude gp130-binding variants, thereby enriching variants with complex specificity and gp130-competitive activity^{8,9}. Furthermore, we expressed the evolved designed protein STAR6 (Synthetic Targeting Agent for Recognition of the IL-6/sIL-6R complex) on the surface of macrophages, extending its function from competitive blockade of the complex to cell-mediated capture and clearance, and thereby generated engineered macrophages. Collectively, this study applies the canonical immunological mechanism of B cell somatic hypermutation to the functional evolution of *de novo*-designed proteins and further translates selective recognition of a cytokine-receptor complex into a cell-mediated clearance function, providing a new strategy for the selective modulation of IL-6 trans-signaling (Extended Data Fig.1).

Result 1. Functional enrichment of the B cell display library and identification of top candidates

We sought to design a *de novo* protein that selectively binds the IL-6/sIL-6R α complex and blocks gp130 engagement (Fig. 1a). The protein comprises three segments: A targets the gp130-binding interface on IL-6; B targets IL-6R α Trp249, which participates in gp130 binding; and C targets Trp296 in the membrane-proximal region of IL-6R α .

Using the crystal structure of the IL-6/sIL-6R α /gp130 complex (PDB ID: 1P9M), we defined three target epitopes and corresponding hotspot constraints. RFdiffusion generated 400

backbones per site across 40 batches, yielding 1,200 backbones in total. ProteinMPNN generated 10 sequences per backbone, which were ranked by their mean per-residue scores; approximately 78 candidates were retained (Extended Data Table 1). AlphaFold-predicted complexes were filtered using a target-aligned binder RMSD < 10 Å and either a binder pTM > 0.7 or a minimum interchain PAE < 5 Å. The three highest-ranking designs per site yielded nine segment designs (Fig. 1b), which were combinatorially assembled into 27 full-length sequences (Extended Data Table 2).

Each design was fused to the FcR γ transmembrane and cytoplasmic domains to enable surface display and receptor-mediated internalization, and the lentiviral library was transduced into Raji cells and puromycin-selected to establish a stable library, with surface expression confirmed by flow cytometry (Fig. 1c).

Library-transduced cells were incubated with the biotinylated IL-6/sIL-6R α complex and stained with streptavidin-APC (Extended Data Fig. 2). A membrane-impermeable fluorescence quencher was then applied to suppress surface-associated fluorescence, allowing retained intracellular APC fluorescence to serve as a readout of complex internalization. Cells were isolated by fluorescence-activated cell sorting according to this signal.

Following the final sorting round, full-length inserts were amplified from genomic DNA using primers targeting conserved vector regions and subjected to long-read sequencing. Of 19,606 valid reads, 14,894 contained complete sequences, corresponding to 75.97% of valid reads (Extended Data Tables 3-4).

A2B3C3 was the most abundant variant, with 2,530 reads (17.0% of full-length sequences), followed by A2B3C2 with 1,157 reads (7.8%) and A3B2C2 with 1,012 reads (6.8%). Together, these variants accounted for 31.6% of full-length sequences after screening. Each exceeded the 3.7% frequency expected under equal representation of the 27 variants, identifying them as the top candidates from the functional screen (Fig. 1d). We also evaluated the top three variants using AlphaFold3 (AF3) structural modeling. AF3 predictions showed that A2B3C3 exhibited the largest buried surface area and formed hydrophobic contacts with both IL-6 and sIL-6R α (Extended Data Tables 5–7).

Result 2. Structural interface and molecular dynamics analyses identify A2B3C3 as the starting sequence for directed evolution

To further assess the binding of the candidates to the IL-6/sIL-6R α complex, we first performed static interface analyses of representative optimized structures for each system using Rosetta. A2B3C3 exhibited the most favorable interface energy (-16.285 Rosetta energy units [REU]) and the largest buried interface area ($2,423.6 \text{ \AA}^2$), indicating that it formed an extensive and energetically favorable binding interface with the IL-6/sIL-6R α complex (Fig. 2a-c). Although A3B2C2 had a relatively smaller interface area, it exhibited the most favorable interface energy density (-0.958) and the highest interface packing score (0.459), indicating a more compact interface with interactions concentrated within a smaller contact area (Fig. 2d-f). The static interface metrics of A2B3C2 were generally intermediate: its predicted interface energy was slightly more favorable than that of gp130, whereas its interface packing was comparable to that of gp130.

To determine whether these static binding conformations could be maintained in solution, we performed 100-ns molecular dynamics (MD) simulations of the three candidate complexes and the gp130 reference complex. A2B3C3 maintained an extensive and stable binding interface throughout the simulation, exhibiting the lowest receptor-side root-mean-square deviation (RMSD; $1.34 \pm 0.24 \text{ \AA}$) and the largest mean buried surface area ($31.82 \pm 1.73 \text{ nm}^2$) (Fig. 2g-j, Extended Data Fig. 3). Radar-plot analysis further showed relatively limited fluctuations in its interfacial van der Waals and electrostatic energies and in residue B factors, indicating that its overall conformation remained well constrained (Fig. 2m).

A3B2C2 had the lowest candidate-protein RMSD ($2.23 \pm 0.55 \text{ \AA}$) and the highest number of interfacial hydrogen bonds (13.34 ± 2.60), consistent with stabilization through a tightly organized polar interaction network despite its smaller interface (Fig. 2g-l). A2B3C2 underwent greater candidate-side conformational rearrangement but retained its interface during the later simulation stages. Its buried surface area and hydrogen-bond retention over the final 20 ns were slightly higher than those of gp130.

Overall, the Rosetta static interface analyses and MD simulations yielded concordant results. The resulting computational prioritization was A2B3C3 (candidate 1) > A3B2C2 (candidate 3) > A2B3C2 (candidate 2) > the gp130 reference complex (Fig. 2; Extended Data Table 8). On the basis of its sequencing enrichment and overall performance in the MD-based evaluation, A2B3C3 was selected as the starting sequence (TOP1) for the second round of directed evolution and screening.

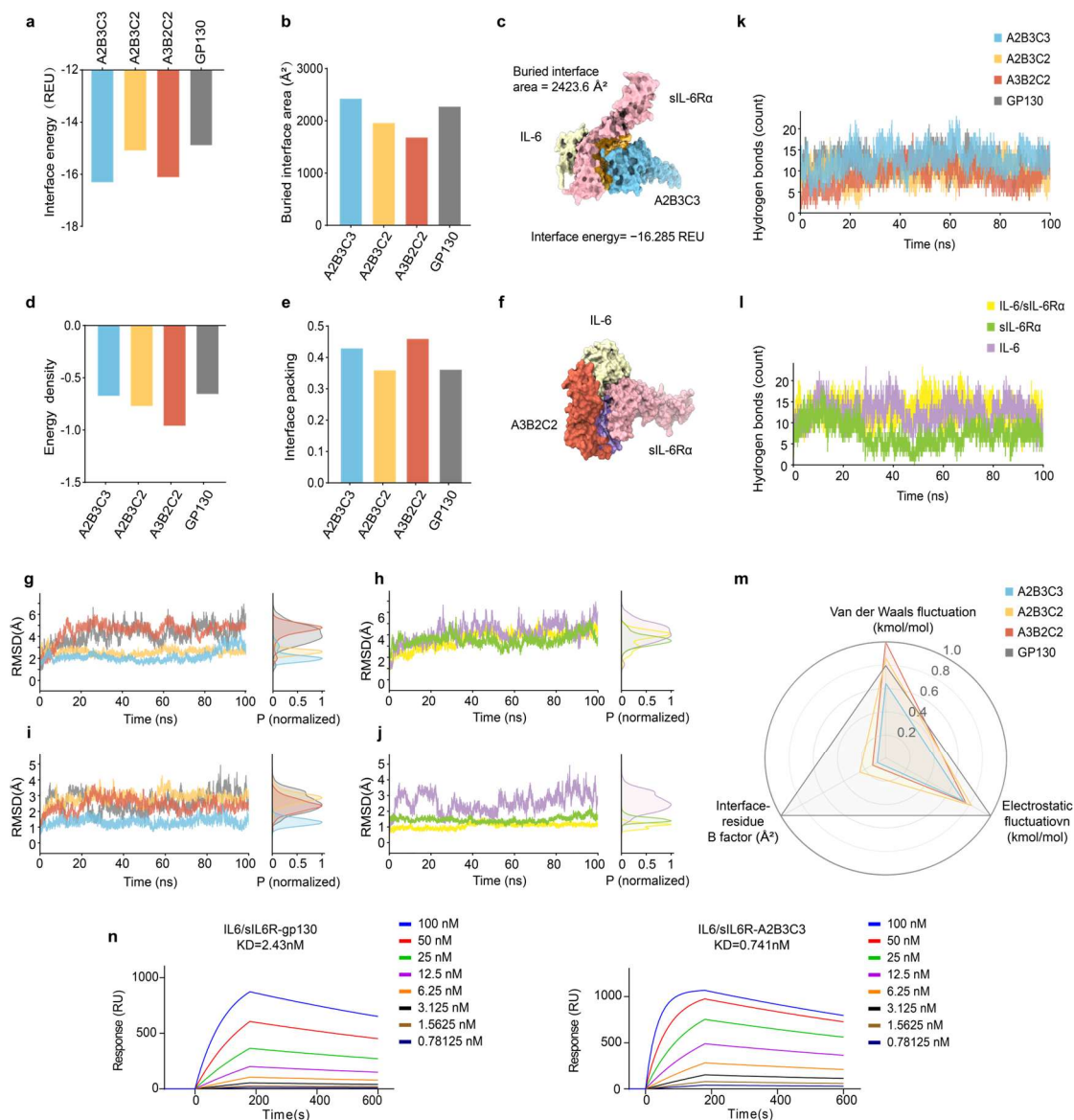


Fig. 2 | Structural, molecular-dynamics and biophysical characterization of the first-round candidate binders. **a, b**, Rosetta-derived interface energy (a) and buried interface area (b) of representative optimized models of A2B3C3, A2B3C2 and A3B2C2, with native gp130 included as the reference. **c**, Structural model of A2B3C3 bound to the IL-6/sIL-6Rα complex,

showing its buried interface area and Rosetta interface energy. **d, e**, Interface energy density (d) and interface packing score (e) of the three candidate binders and gp130. **f**, Structural model of A3B2C2 bound to the IL-6/sIL-6R α complex, illustrating its compact binding interface. **g-j**, Root-mean-square deviation (RMSD) trajectories and corresponding normalized probability distributions obtained from 100-ns molecular-dynamics simulations. Candidate-protein or gp130 RMSDs are shown in **g**, and IL-6/sIL-6R α receptor-side RMSDs are shown in **i**; component-resolved RMSD analyses of the A2B3C3-bound system are shown in **h** and **j**. **k**, Number of interfacial hydrogen bonds formed during the 100-ns simulations of the three candidate complexes and the gp130 reference complex. **l**, Component-resolved hydrogen-bond analysis of A2B3C3 with the complete IL-6/sIL-6R α complex or with the individual IL-6 and sIL-6R α components. **m**, Radar plot comparing the normalized fluctuations in interfacial van der Waals energy, electrostatic energy and interface-residue B factors. **n**, SPR sensorgrams comparing the interactions of the IL-6/sIL-6R α complex with gp130 or A2B3C3 at the indicated concentrations. The complete 100-ns trajectories are displayed; the first 20 ns were treated as structural relaxation, and the values reported in the text were calculated over 20-100 ns.

Result 3. dCas9-AID* Δ -driven directed evolution of A2B3C3 yields a potent IL-6/sIL-6R α -capturing molecule

Building on A2B3C3, which was identified through computational design and B cell surface-display screening, we initiated directed evolution to further optimize this lead sequence. Localized point mutations were introduced into the A2, B3, and C3 coding regions, and binding to and uptake of the IL-6/sIL-6R α complex were used as selection pressures to iteratively enrich full-length variants with enhanced complex-capture capacity, thereby improving their ability to compete with gp130.

To enable programmable hypermutation of A2B3C3 in Raji B cells, we developed dCas9-AID* Δ by combining design features of CRISPR-X and targeted AID-mediated mutagenesis (TAM). CRISPR-X uses sgRNAs containing two MS2 aptamers to recruit MS2-AID fusion proteins. Its hyperactive AID* Δ variant incorporates three gain-of-function substitutions (K10E,

T82I, and E156G) and deletion of the three C-terminal residues⁸. TAM instead directly fuses a P182X-truncated AID variant, termed AIDx, to dCas9 to target mutagenesis to sgRNA-defined regions⁹. Following this direct fusion architecture, we fused the hyperactive AID*Δ variant to dCas9 to generate dCas9-AID*Δ, coupling programmable DNA targeting with enhanced AID activity (Fig. 3b).

For the second round of directed evolution, full-length A2B3C3 identified in the first-round screen was used as a single evolutionary unit. The A2, B3, and C3 binding segments were assembled in frame by Golden Gate cloning to generate a continuous open reading frame, which was subsequently linked to the transmembrane and intracellular signaling modules for surface display in Raji B cells (Fig. 3a)¹⁰. sgRNAs were designed to target the A2, B3, and C3 coding regions and direct dCas9-AID*Δ-mediated sequence diversification to the three corresponding regions. Thus, rather than generating predetermined substitutions at individual nucleotide positions, the three sgRNAs established three locally enriched mutational windows within the same full-length A2B3C3 coding sequence. Following activation of the targeted mutagenesis system in Raji B cells, the cells were incubated with a fluorescently labeled IL-6/sIL-6Rα complex and subjected to flow cytometric sorting based on retained complex-associated fluorescence.

To characterize AID*Δ-generated mutational profiles, PacBio HiFi full-length reads were aligned to the Golden Gate-assembled A2B3C3 reference sequence, and nucleotide substitution frequencies were calculated at each position¹¹. Using an operational threshold of 0.1%, substitutions were preferentially enriched within the three sgRNA-targeted regions in A2, B3, and C3 and their flanking sequences, whereas most distal positions remained at or below this threshold (Fig. 3c). Substitutions included G>A, C>T, G>C, C>A, and C>G across the targeted regions.

To further characterize the nucleotide preference of the induced mutations, substitution events detected within the A2, B3, and C3 regions were classified according to the corresponding reference nucleotide. Substitutions originating from C or G accounted for 54.7%, 68.6%, and 65.0% of all substitutions detected in the A2, B3, and C3 regions, respectively (Fig. 3d), indicating that the mutations predominantly originated from C:G base pairs. Because AID

directly catalyzes cytosine deamination in single-stranded DNA, the resulting mutation may be recorded as either a C- or G-derived substitution depending on the orientation of the reference strand. This distribution was therefore consistent with the biochemical mechanism of AID-mediated cytosine deamination.

The substitution outcomes of mutations originating from C, G, A, or T were subsequently analyzed separately. In addition to the canonical C>T and complementary G>A transitions, the dCas9-AID*Δ system generated a broader spectrum of nucleotide substitutions through subsequent DNA replication and mismatch-repair processes. The resulting candidate sequences were prioritized for variant reconstruction and further evaluation of IL-6/sIL-6Rα binding and cellular internalization (Fig. 3c).

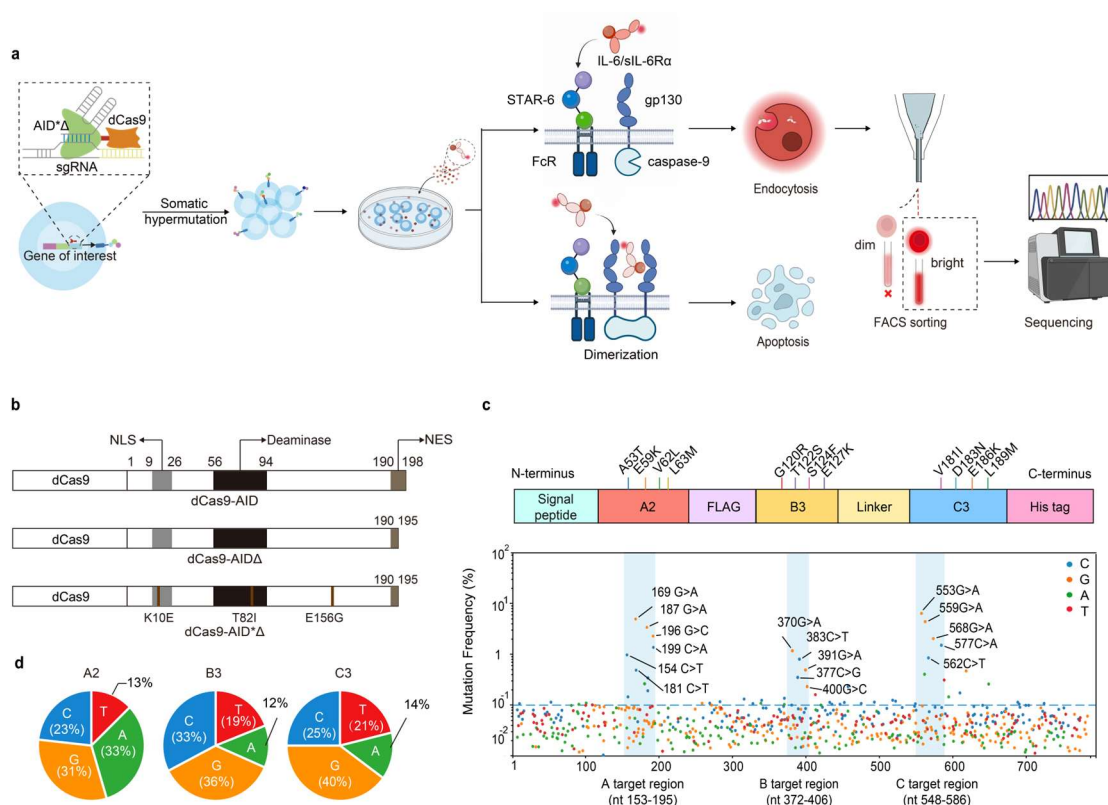


Fig. 3 | Localized mutation distribution and nucleotide-substitution spectrum across the full-length A2B3C3 coding sequence. **a**, Schematic of the second-round directed-evolution and sequencing workflow. The A2, B3 and C3 segments were assembled into a continuous open reading frame, and the experimental group was established using sgRNAs targeting the A2, B3 or C3 region. Following fluorescence-based sorting with the IL-6/sIL-6Rα complex, an

amplicon spanning all three segments was subjected to PacBio HiFi full-length sequencing. **b**, Schematic of dCas9-AID variants and the final expression cassette. Domain organization of dCas9-AID, dCas9-AID Δ , and dCas9-AID* Δ . The NLS (aa 9-26), deaminase domain (aa 56-94), and NES (aa 190-198) are indicated. Compared with dCas9-AID, dCas9-AID Δ lacks the C-terminal three amino acids. dCas9-AID* Δ is based on dCas9-AID and additionally carries the K10E, T82I, and E156G mutations. The NLS, deaminase domain, and NES are retained in these constructs. **c**, Representative high-frequency mutations within the A2, B3 and C3 regions. Per-nucleotide substitution frequencies relative to the 789-nt reference sequence. Position 1 of the reference sequence was defined as the origin of the x axis, and the start codon of the coding region was located at nt 13. The substitution frequency at each position was calculated as the number of high-quality reads containing a nucleotide substitution divided by the total number of reads covering that position. Colors indicate the corresponding reference nucleotide, and the dashed line denotes the 0.1% frequency threshold used to define detectable mutated sites. The principal mutations in group A included G169>A (24.44%), G187>A (15.38%), G196>C (8.76%) and C199>A (4.99%); those in group B included G370>A (19.20%), C383>T (13.07%), G391>A (7.95%) and C377>G (5.67%); and those in group C included G553>A (24.68%), G559>A (16.98%), G568>A (9.21%) and C577>A (7.08%). All nucleotide positions are numbered according to the same 789-nt reference sequence. **d**, Composition of detectable mutated sites in the A2-, B3- and C3-targeted groups, classified according to the reference nucleotide (A, T, C or G).

Result 4. Second-round evolution enhances interfacial interactions between candidate proteins and the IL-6/sIL-6R α complex

To further investigate the structural basis underlying the stable association of the evolved candidates with the IL-6/sIL-6R α complex, we used AlphaFold 3 to generate structural models of the corresponding protein complexes (Fig. 4a-c). Computational alanine scanning was subsequently performed to identify hotspot residues at the predicted binding interfaces (Fig. 4d; Extended Data Table 12). All three evolved candidates exhibited substantially increased buried surface areas relative to the first-round candidates. Among them, S2 formed the largest buried

interface, with a buried surface area of 2,146.193 Å², compared with 1,744.917 Å² for S1 and 1,423.955 Å² for S3. Although the three candidates showed no appreciable difference in their propensity to form hydrogen bonds, S2 formed a greater number of hydrophobic contacts (Extended Data Table 9-11), which may contribute to the stabilization of its bound conformation.

Separate analysis of the candidate interfaces with IL-6 and sIL-6R α revealed that all three candidates adopted a relatively conserved interaction pattern at the IL-6-binding interface, involving the Tyr13-Arg6-Leu1 interaction network. By contrast, the C-terminal helical region of S2 inserted more deeply into the sIL-6R α interface, resulting in a marked increase in contacts with hotspot residues. Among these residues, sIL-6R α Trp169 exhibited the highest contact count (1,468) and was identified as the predominant hotspot residue at the interface.

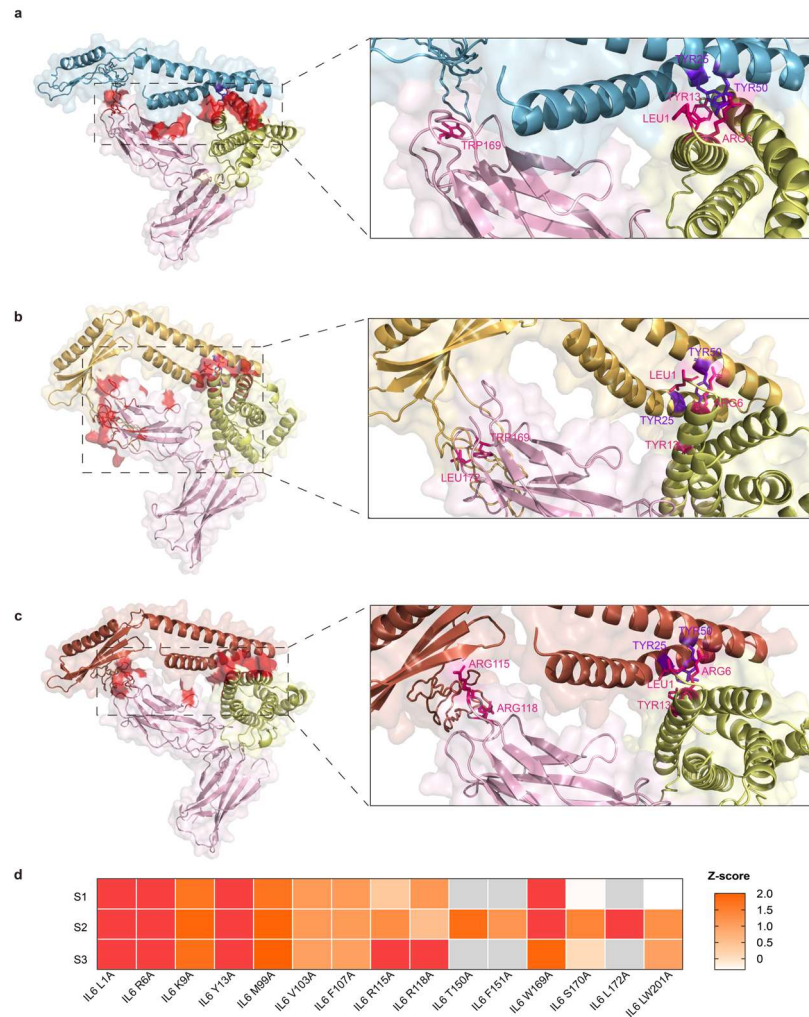


Fig. 4 |In silico construction of second-round evolution candidates. a-c, Ninety repeated predictions of S1, S2, and S3 were performed using AlphaFold3. The five predictions with the

highest ipTM scores were selected as high-confidence models, and the averaged virtual complex structures derived from their superposition (left), together with the hot-spot residues identified by alanine scanning ($Z\text{-score} > 2$) (right), are shown. S1, S2, and S3 are shown in cyan, gold, and salmon, respectively; IL-6 and sIL-6R α are shown in yellow and pink, respectively. Hot-spot residues on the IL-6/sIL-6R α complex are highlighted in magenta, and those on the candidates are highlighted in purple. The binding interface is highlighted in red in the left panels. **d**, Heatmap of Z-scores for hot-spot residues on the IL-6/sIL-6R complex side. The vertical axis represents the three candidates, and the horizontal axis corresponds to alanine mutations of the hot-spot residues. The white-to-orange gradient indicates increasing Z-scores, red denotes hot-spot residues with $Z\text{-scores} > 2$, and grey indicates residues absent from the binding interface of the corresponding complex.

Result 5. STAR6 binds the IL-6/sIL-6R α complex with picomolar affinity

We designated the selected candidate S2 as STAR6. To characterize the binding properties of STAR6, we measured its affinity for the IL-6/sIL-6R α complex using SPR. As the concentration of the IL-6/sIL-6R α complex increased from 15.625 to 2,000 pM, the SPR response of STAR increased in a concentration-dependent manner. Kinetic analysis yielded an equilibrium dissociation constant (K_D) of approximately 52 pM, demonstrating that STAR bound the IL-6/sIL-6R α complex with high affinity (Fig. 5a). Thus, directed evolution improved the affinity of STAR6 for the IL-6/sIL-6R α complex by approximately 14.3-fold relative to A2B3C3 (Fig. 2n). Having established that STAR6 recognized the IL-6/sIL-6R α complex with high affinity and competitively inhibited its interaction with gp130, we next displayed STAR6 on the surface of macrophages as an extracellular recognition module coupled to the cytoplasmic domain of FcR γ (Fig. 5b). We termed these engineered macrophages STAR-Trap to reflect their intended ability to capture and clear the complex through receptor-mediated internalization. Stable establishment of the STAR-Trap population and surface expression of the STAR6–FcR γ receptor were confirmed by flow cytometry (Fig. 5c).

To assess STAR-Trap-mediated internalization of the IL-6/sIL-6R α complex, STAR-Trap were incubated with biotinylated IL-6/sIL-6R α complexes fluorescently labeled with

streptavidin-APC. Extracellular fluorescence was then quenched, leaving only intracellular signals detectable. Flow cytometry revealed a higher proportion of fluorescence-positive cells and greater mean intracellular fluorescence intensity in STAR-Trap cells than in control macrophages, indicating enhanced internalization of the complex (Fig. 5d).

Confocal fluorescence imaging provided further spatial evidence for the intracellular fate of the captured complex. STAR-Trap exhibited prominent intracellular signals derived from the fluorescent IL-6/sIL-6R α complex probe, whereas little probe fluorescence was detected in control macrophages (Fig. 5e). Moreover, the internalized complex showed substantial colocalization with LysoTracker-positive compartments, indicating that STAR-Trap-mediated uptake was followed by trafficking of the complex to lysosomes. Together, these findings demonstrate that STAR6 combines high-affinity recognition of the IL-6/sIL-6R α complex with competitive inhibition of its interaction with gp130. Incorporation of STAR6 into an FcR γ -containing macrophage surface receptor equipped STAR-Trap cells with the capacity to capture, internalize and deliver the complex to lysosomal compartments.

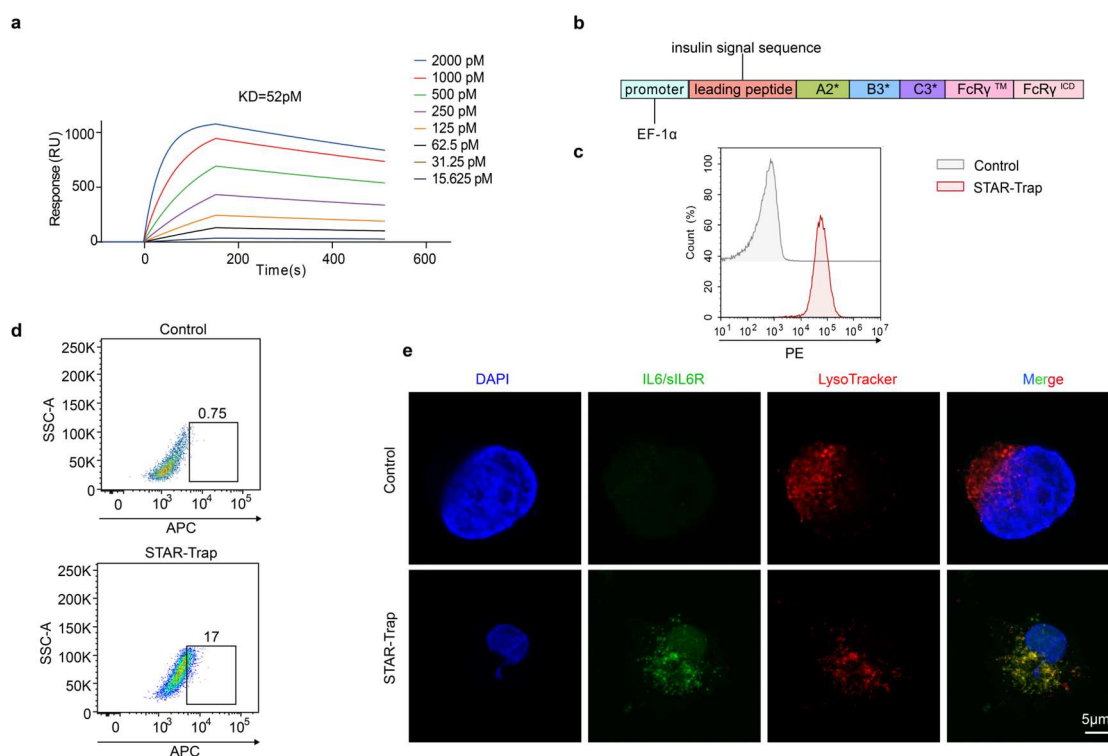


Fig. 5 | STAR6 binds the IL-6/sIL-6R α complex with picomolar affinity and enables its macrophage-mediated internalization and lysosomal trafficking. **a**, Representative surface plasmon resonance (SPR) sensorgrams showing the binding of STAR to the IL-6/sIL-6R α complex at the indicated concentrations (15.625-2,000 pM). Kinetic fitting yielded an

equilibrium dissociation constant (K_D) of 52 pM. **b**, Schematic of the lentiviral expression cassette encoding the STAR-6-FcR γ receptor used to generate STAR-Trap. The evolved A2*, B3* and C3* segments constitute the extracellular STAR-6 recognition domain, which was placed downstream of an insulin signal peptide and fused to the transmembrane (TM) and intracellular signaling domains (ICD) of FcR γ under the control of the EF-1 α promoter. **c**, Representative flow-cytometry histograms confirming the establishment of a stable STAR-Trap population and surface expression of the STAR6-FcR γ receptor. Grey, control macrophages; red, STAR-Trap. **d**, Representative flow-cytometry plots showing uptake of the biotinylated IL-6/sIL-6R α complex labeled with streptavidin-APC by control macrophages (Control) and STAR-Trap after quenching extracellular fluorescence. Numbers indicate the percentages of APC-positive cells within the indicated gates. **e**, Confocal fluorescence images showing the intracellular distribution of the fluorescent IL-6/sIL-6R α complex probe and its colocalization with lysosomes in control and STAR-Trap. Nuclei were stained with DAPI (blue), the IL-6/sIL-6R α complex probe is shown in green, and lysosomes were labeled with LysoTracker (red). Scale bars, 5 μ m.

Result 6. STAR-Trap engineering biases macrophages toward an M1-like phenotype

We next examined whether STAR-Trap engineering altered the polarization state of macrophages. Flow cytometry was used to evaluate the M1-associated markers HLA-DR and CD86, intracellular reactive oxygen species (ROS), and the M2-associated markers CD206, Arg-1, and IL-10 (Fig. 6a-b). Compared with control macrophages, STAR-Trap cells exhibited increased HLA-DR and CD86 expression together with elevated intracellular ROS levels. In contrast, CD206 and IL-10 levels were reduced, whereas Arg-1 showed only a minimal change. Collectively, the coordinated increase in M1-associated features and reduction in several M2-associated markers indicate that STAR-Trap engineering was associated with a shift toward an M1-like macrophage phenotype.

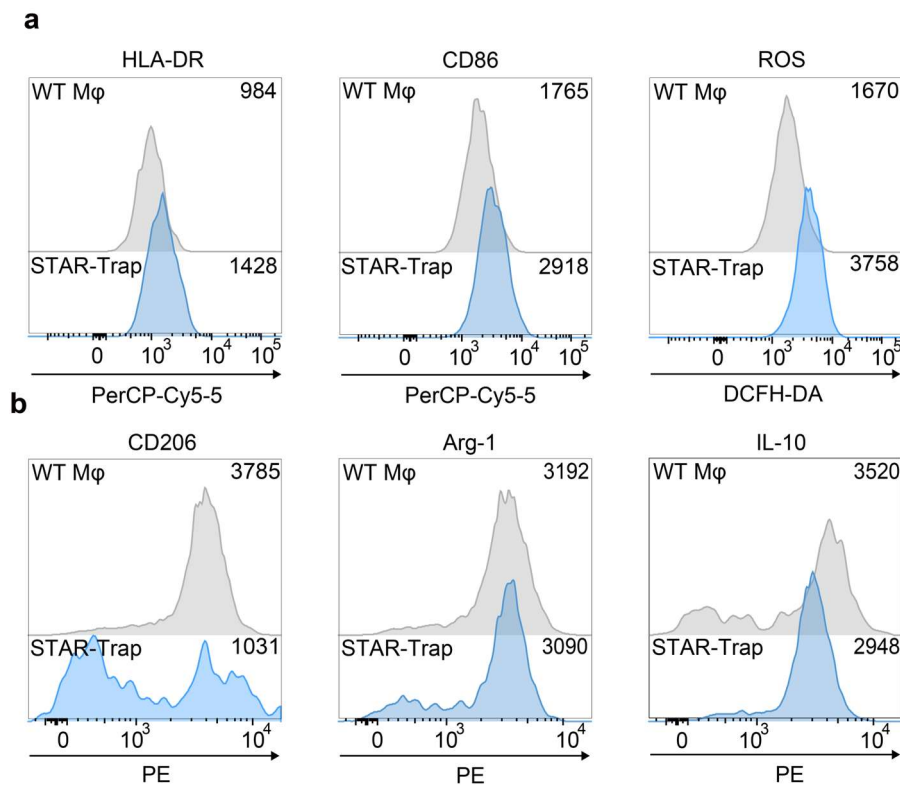


Fig. 6 | STAR-Trap exhibit an M1-like phenotypic shift. **a**, Representative flow-cytometry histograms showing the expression of the M1-associated markers HLA-DR and CD86 and intracellular ROS levels in wild-type macrophages (WT Mφ) and STAR-Trap. **b**, Representative flow-cytometry histograms showing the expression of the M2-associated markers CD206, Arg-1 and IL-10 in WT and STAR-Trap. Grey histograms represent WT macrophages, and blue histograms represent STAR-Trap. Numbers indicate the median fluorescence intensity of the corresponding cell populations.

Result 7. STAR-Trap selectively suppresses IL-6 trans-signaling while preserving classical signaling

To determine whether STAR-Trap selectively inhibited IL-6 trans-signaling while preserving classical signaling, we established two separate cell-based assays to evaluate classical signaling and trans-signaling, respectively. To assess IL-6 trans-signaling, the IL-6/sIL-6Rα complex was preincubated with wild-type macrophages or STAR-Trap. After removal of the macrophages, the resulting supernatants were applied to human umbilical vein endothelial cells (HUVECs). The IL-6/sIL-6Rα complex induced robust STAT3 phosphorylation and increased MCP-1 and ICAM-1 expression in HUVECs (Fig. 7a). These

responses remained pronounced in HUVECs treated with supernatants collected after preincubation of the complex with wild-type macrophages, but were markedly reduced in HUVECs treated with supernatants from the STAR-Trap condition, as demonstrated by decreased intracellular pSTAT3 levels and reduced MCP-1 and ICAM-1 expression.

To assess classical signaling, free IL-6 was preincubated with either wild-type macrophages or STAR-Trap. Cell-free supernatants were subsequently collected and applied to U266 human multiple myeloma cells, which express both membrane-bound IL-6R α (mIL-6R α) and gp130. Free IL-6 induced robust STAT3 and ERK1/2 phosphorylation together with increased SOCS3 expression in U266 cells (Fig. 7b). These responses remained largely unchanged when U266 cells were treated with supernatants collected after preincubation of free IL-6 with either wild-type or STAR-Trap. By contrast, treatment with the IL-6R-blocking antibody tocilizumab substantially reduced pSTAT3, pERK1/2, and SOCS3 levels in U266 cells. Thus, STAR-Trap did not measurably impair classical signaling mediated by free IL-6.

Together, these findings demonstrate that STAR-Trap selectively removes the IL-6/sIL-6R α complex and suppresses the associated trans-signaling response while largely preserving classical signaling mediated by free IL-6.

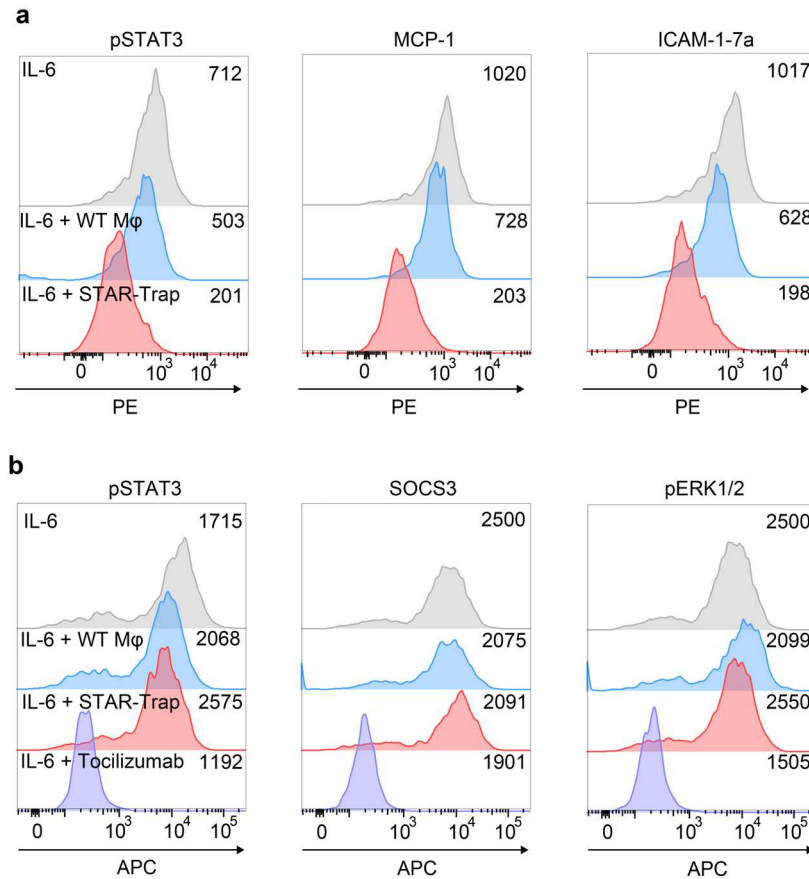


Fig. 7 | STAR-Trap attenuates IL-6 trans-signaling while preserving classical signaling.

a, Representative flow-cytometry histograms showing IL-6 trans-signaling responses. The IL-6/sIL-6R α complex was incubated alone or preincubated with WT or STAR-Trap. After removal of the macrophages, the resulting cell-free supernatants were applied to gp130-positive responder cells with low or undetectable mIL-6R α expression. Intracellular pSTAT3 and the downstream inflammatory markers MCP-1 and ICAM-1 were subsequently measured. Grey, IL-6/sIL-6R α complex alone; blue, complex preincubated with WT macrophages; red, complex preincubated with STAR-Trap. **b**, Representative flow-cytometry histograms showing classical IL-6 signaling responses. Free IL-6 was incubated alone or preincubated with WT or STAR-Trap, and the resulting cell-free supernatants were applied to responder cells expressing both mIL-6R α and gp130. Intracellular pSTAT3, SOCS3 and pERK1/2 were measured, with tocilizumab included as an IL-6R-blocking control. Grey, IL-6 alone; blue, IL-6 preincubated with WT macrophages; red, IL-6 preincubated with STAR-Trap; purple, tocilizumab-treated control. Numbers indicate the median fluorescence intensity of the corresponding cell populations.

Discussion

Tocilizumab and other IL-6R-blocking antibodies, such as sarilumab, are anti-inflammatory biologics used clinically for conditions including moderate-to-severe rheumatoid arthritis. By blocking IL-6R, these agents inhibit both classical and trans-signaling, potentially increasing the risk of immunosuppression and infection¹². By contrast, selective recognition of the IL-6/sIL-6R α complex may restrict trans-signaling while preserving mIL-6R-dependent classical signaling. On this basis, STAR6 was designed to recognize a composite epitope formed only upon assembly of IL-6 with sIL-6R α , thereby avoiding direct binding to free IL-6 or membrane-bound IL-6R. This design is conceptually similar to sgp130Fc (olamkicept), which selectively inhibits IL-6 trans-signaling, although the two molecules act through distinct mechanisms¹³. STAR-Trap functions as a *living scavenger*. Macrophages can naturally home to inflamed tissues and, as professional phagocytes, possess the lysosomal machinery needed to process captured complexes efficiently¹⁴. They also participate in immune regulation, so changes in their phenotype may affect neighboring immune cells¹⁵. We considered mesenchymal stromal cells (MSCs) as an alternative cellular platform because they also have inflammation-homing and immunomodulatory properties. However, MSCs predominantly express the inhibitory Fc γ RIIB receptor and lack robust internalization signaling mediated by canonical activating Fc γ Rs. Their Syk-, PI3K- and actin-remodeling-associated pathways are comparatively limited, thereby constraining FcR-dependent uptake of protein complexes and their subsequent lysosomal processing. Consequently, MSCs may be less capable of efficiently sustaining the capture, internalization and degradation of the IL-6/sIL-6R α complex and are therefore less well suited to completing the full process of target clearance¹⁶⁻¹⁸.

In selecting an AID-based mutagenesis system, we weighed mutation rate against mutational bias. In CRISPR-X, dCas9 recruits four AID molecules through MS2 hairpins; the AID variant also carries three point mutations. This increases the local concentration of AID and raises the mutation rate¹⁹. TAM instead uses a direct dCas9-AID fusion. Binding of dCas9-AID to the target DNA creates steric hindrance that impedes replication-fork progression and prevents DNA polymerase from interpreting uracil as thymine. Uracil is consequently directed toward replication-independent repair pathways, reducing mutational bias²⁰. This system was

designed to combine the advantages of both approaches by incorporating the three AID substitutions used in CRISPR-X into the TAM architecture, with the aim of achieving a higher mutagenesis efficiency while reducing C/G mutational bias. This strategy was expected to broaden the spectrum of accessible amino acid substitutions and expand the sequence space available for identifying beneficial mutations.

Our design has several limitations. First, although we detected no *in vitro* binding of our protein to free IL-6 or sIL-6R α , off-target effects in complex biological fluids cannot be excluded. Second, the observed M1 polarization is a pro-inflammatory phenotype, in contrast to the anti-inflammatory and tissue-repairing M2 phenotype; while M1 polarization might aid inflammatory clearance, it could also aggravate tissue injury if left uncontrolled. Future studies should incorporate a safety switch, such as inducible caspase9, to permit elimination of the engineered cells when necessary.

Materials and Methods

Cell culture

The human Burkitt lymphoma B cell line Raji and the human monocytic leukemia cell line THP-1 were cultured in RPMI-1640 medium supplemented with 10% fetal bovine serum (FBS). The human multiple myeloma cell line U266 was maintained in RPMI-1640 medium supplemented with 15% FBS. HUVECs were cultured in complete endothelial cell growth medium. All cells were maintained at 37 °C in a humidified atmosphere containing 5% CO₂.

Computational design

The crystal structure of the IL-6/sIL-6R α /gp130 ternary complex (PDB ID: 1P9M) was used as the structural template. The binding interfaces were examined in detail using PyMOL, leading to the identification of three functional epitope regions: Part A, corresponding to the IL-6-gp130 interface; Part B, corresponding to the gp130-binding site centered on Trp249 of IL-6R α ; and Part C, corresponding to the Trp296 site within the membrane-proximal region of IL-6R α . Key residues closely involved in binding within each epitope were defined as hotspot residues and

used as spatial constraints for subsequent backbone generation. In parallel, the signaling-complex interfaces of other IL-6 family cytokines were analyzed by integrating published structural data with PyMOL-based structural comparisons to exclude potential hotspots associated with nonspecific binding.

Binders A, B, and C were designed to specifically recognize the three epitopes described above. Binder A was designed to target the Leu19/Tyr31 interface of IL-6 (Part A), binder B to target the interface centered on Trp249 of IL-6R α (Part B), and binder C to target the interface centered on Trp296 of IL-6R α (Part C). To obtain a diverse set of initial backbones, RFdiffusion was used to independently generate backbones for each target. Each generation run was initialized from random noise and subjected to an iterative denoising process to produce a complete protein backbone conformation. Different random seeds were used across generation runs to enhance conformational diversity. For each target, 40 independent batches were generated, with each batch producing 10 distinct backbone conformations, resulting in 400 initial backbones per target.

ProteinMPNN was subsequently used to design amino acid sequences for all generated backbones. During sequence design, the sequence and structure of the target were kept fixed, and only the amino acid sequence of the binder was optimized. For each backbone conformation, ProteinMPNN generated 10 candidate sequences. These sequences were ranked according to their mean per-residue ProteinMPNN scores, and the top-ranked sequence for each backbone was retained for subsequent virtual screening.

The structures of the candidate protein-target complexes were predicted using AlphaFold, including models of the designed proteins in complex with IL-6/sIL-6R α . Virtual screening was performed using three metrics: the predicted template modeling score of the binder-target complex (binder pTM), the minimum interchain predicted aligned error (PAE), and the binding-site root-mean-square deviation (RMSD) calculated after alignment to the target. Designs with a target-aligned binder RMSD < 10 Å and either a binder pTM > 0.7 or a minimum interchain PAE < 5 Å were retained for further evaluation.

The retained candidates were further evaluated using Rosetta-based statistical energy

calculations (ddG) and 50-ns molecular dynamics simulations performed with GROMACS. Candidates with a ddG < -10 REU and stable RMSD profiles throughout the simulation trajectory were retained, yielding 79 candidate sequences. Following integrated evaluation of binder pTM, PAE, RMSD, and MD-RMSD, three prioritized sequences were selected for each binding site, resulting in nine final candidate segment sequences: A1-A3, B1-B3, and C1-C3.

Construction of the combinatorial library by homologous recombination

The nine candidate segment sequences were combinatorially assembled into the pDisplay vector linearized with XbaI and BamHI using a homologous recombination kit (Vazyme). Overlapping regions of 20-40 bp were introduced at both ends of each insert to enable homologous assembly of the A, B, and C segments.

The 5' end of each Part A fragment contained a sequence homologous to the left end of the linearized vector (AGCTGTGACCGGCGCCTACTCTAGA), followed by the signal peptide-coding sequence (ATGGCCCTCTGGATGAGGCTGCTGCCCCCTGCTGGCATTGCTGGCCCTGTGGGGCCCTGATCCCGCTGCCGCA). Its 3' end contained a FLAG tag-coding sequence (GACTACAAGGACGACGACGACAAG), which served as the homologous overlap for assembly with the Part B fragment.

The 5' end of each Part B fragment contained the same FLAG tag-coding sequence, whereas its 3' end contained a linker-coding sequence (GGCGGGGGCAGCGGCGGGGGCTCC), which served as the homologous overlap for assembly with the Part C fragment. The 5' end of each Part C fragment contained the same linker-coding sequence, whereas its 3' end contained a sequence homologous to the FcR transmembrane region of the vector (ATCATTGTGGCGGTGGTGAT). The assembled products were transformed into Fast-T1 competent cells, after which plasmid DNA was isolated.

Lentiviral packaging and construction of the B cell library

The plasmid library encoding the 27 *de novo* designs generated by homologous recombination was mixed with the packaging plasmid pCMVR8.74 and the envelope plasmid pMD2.G at a mass ratio of 3:2:1, with a total DNA amount of 30 µg. HEK293T cells cultured in 10-cm dishes to approximately 75% confluence were transfected using linear polyethylenimine with a

molecular weight of 25,000 (PEI; PEI:DNA mass ratio, 3:1). At 6 h after transfection, the medium was replaced with fresh complete culture medium. Cell-culture supernatants containing lentiviral particles were collected at 48 and 72 h after transfection, pooled, and passed through a 0.45- μ m filter to remove cellular debris. The viral particles were concentrated by ultracentrifugation at $20,000 \times g$ for 90 min at 4°C. The resulting viral pellet was resuspended in PBS or complete culture medium, aliquoted, and stored at -80°C until use.

Exponentially growing Raji B cells, a human B cell lymphoma cell line, were transduced with the concentrated lentivirus at a multiplicity of infection (MOI) of 0.3. Polybrene was added at a final concentration of 8 μ g/mL to enhance transduction efficiency. The cells were subjected to spinoculation at 1,500 rpm for 1.5 h at room temperature and subsequently cultured at 37°C in a humidified incubator containing 5% CO₂. Puromycin selection was initiated 72 h after transduction. The appropriate puromycin concentration was determined in a preliminary titration experiment. Untransduced Raji B cells were treated with 0, 0.5, 1, 2, or 5 μ g/mL puromycin for 5 days, and the lowest concentration that completely eliminated the untransduced cells, 1 μ g/mL in this study, was used for selection. Selection was continued for 5 days, with the medium replaced every 2 days with fresh puromycin-containing medium. Following selection, the cells were expanded for 7 days until a total of 1×10^7 cells was obtained for subsequent fluorescence-activated cell sorting.

Preparation of the fluorescently labeled complex

Recombinantly expressed and purified human sIL-6R and IL-6 were mixed at a 1:1 molar ratio and incubated for 30 min at room temperature to form the sIL-6R/IL-6 complex. The complex was biotinylated using the EZ-Link Sulfo-NHS-Biotinylation Kit (Thermo Scientific, catalog no. 21425). After buffer exchange into PBS, the complex was incubated with a 20-fold molar excess of Sulfo-NHS-Biotin for 30 min at room temperature in the dark, allowing biotin to be covalently conjugated to primary amine groups on the proteins through stable amide bonds. Unreacted free biotin was removed using a Zeba Spin Desalting Column, yielding the purified biotinylated complex. The biotinylated complex was then mixed with streptavidin-APC (BD Biosciences, catalog no. 554067) at a molar ratio of 1:4 and incubated for 30 min at 4°C in the dark to generate the fluorescently labeled sIL-6R/IL-6 complex.

Fluorescence-activated cell sorting

The B cell library was incubated with the fluorescently labeled complex at 37°C for 4 h at a final complex concentration of 50 nM. The cells were subsequently washed twice with ice-cold PBS to terminate internalization and remove unbound fluorescent complexes. Cells were then sorted by flow cytometry on the basis of intracellular APC fluorescence intensity. Sorting gates were established using unstained cells and cells incubated at 4°C, under which internalization was inhibited, as controls. The 5% of cells exhibiting the highest APC fluorescence intensity were collected. Following each round of sorting, the positive cells were expanded for 10 days before the next round of screening. The complex concentration was reduced to 10 nM in the second round and to 2 nM in the third round.

PacBio long-read sequencing

Genomic DNA was extracted from the positive cells obtained after the final round of sorting. Because the combinations of the A, B, and C segments after selection were unknown, whereas the vector-backbone sequence was known and invariant, PCR primers were designed to anneal to constant regions of the vector located upstream of the A segment and downstream of the C segment. These primers were used to amplify the complete insert. Amplicon specificity was verified by agarose gel electrophoresis, after which the corresponding band was excised and purified. The purified amplicons were submitted to a commercial service provider for long-read sequencing on the PacBio platform.

The sequencing data comprised full-length reads spanning the complete insert, including the A, B, and C segments and all intervening junction regions. Because the homologous-arm sequences flanking the segments were known and invariant, they were used as anchor sequences for reading. Specifically, the junction between the A and B segments was delineated by the FLAG tag-coding sequence, which, following assembly, served as both the terminal sequence of the A segment and the initial sequence of the B segment. The junction between the B and C segments was delineated by the GGGGS linker-coding sequence. The boundaries between the complete insert and the vector backbone were defined by the homologous-arm sequences at the left end of the A segment and the right end of the C segment. Using these four

anchor sequences, the start and end positions of each segment were identified within the full-length reads, enabling extraction of the complete nucleotide sequences of the individual A, B, and C segments.

Molecular Dynamics Simulation

Molecular dynamics simulations were performed using GROMACS for four representative Rosetta-refined protein-protein complexes, including candidate 1, candidate 2, candidate 3, and the reference complex. Each complex was parameterized using a protein-compatible force field and placed in a periodic simulation box with sufficient solvent padding between the protein and the box boundaries. The systems were solvated with explicit water molecules, neutralized with counterions, and adjusted to the designated salt concentration. Following energy minimization, the solvated systems were equilibrated sequentially under the constant-number, constant-volume, and constant-temperature (NVT) ensemble and the constant-number, constant-pressure, and constant-temperature (NPT) ensemble. The equilibrated structures were subsequently used as the starting conformations for 100-ns production simulations.

Before trajectory analysis, periodic boundary artifacts were removed, the protein complexes were centered, and the trajectories were fitted to eliminate overall translational and rotational motions. Receptor and ligand root-mean-square deviations (RMSDs) were calculated separately using consistent atom selections to evaluate the conformational stability of the two binding components. The number of interfacial hydrogen bonds and the total buried solvent-accessible surface area (BSA) were calculated to assess the persistence and extent of the binding interfaces. Relative free-energy landscapes were constructed using the first two principal components (PC1 and PC2), with the relative free energy calculated as $\Delta G = -k_B T \ln[P(PC1, PC2)/P_{max}]$. The initial 0-20 ns of each production trajectory was regarded as an additional structural relaxation period and excluded from the primary statistical analysis. Mean values and standard deviations were calculated over the 20-100 ns interval, while the final 80-100 ns interval was additionally examined to identify potential late-stage conformational drift or interface rearrangement.

Construction of the gp130-Caspase9 suicide gene system

To establish the suicide gene system, a fusion gene encoding the extracellular domain of human gp130 and the intracellular effector domain of caspase9 was cloned into the pCDH-EF1 α -MCS-T2A-Puro lentiviral expression vector. The resulting lentiviral vector was co-transfected into HEK293T cells with the packaging plasmid psPAX2 and the envelope plasmid pMD2.G at a mass ratio of 4:3:1 to produce recombinant lentiviral particles. Raji B cells stably expressing the TOP1 designed protein were transduced at a multiplicity of infection (MOI) of 0.3. A stable cell pool co-expressing the TOP1 designed protein and the gp130-Caspase9 fusion protein was subsequently established by fluorescence-activated cell sorting. Cell-surface expression of the fusion protein was confirmed by flow cytometry using an anti-gp130 antibody.

Construction and induction of the dCas9-AID* Δ mutagenesis system

To enable targeted hypermutation of the TOP1 coding sequence, the dCas9-AID* Δ fusion gene was cloned into the pCDH-EF1 α -MCS-T2A-Puro lentiviral expression vector. The resulting vector was co-transfected into HEK293T cells with psPAX2 and pMD2.G at a mass ratio of 4:3:1 to produce recombinant lentiviral particles. Raji B cells stably expressing the TOP1 designed protein were transduced at an MOI of 0.3, and a stable dCas9-AID* Δ -positive cell pool was established by fluorescence-activated cell sorting. The fusion protein comprised catalytically inactive Cas9 (dCas9), which recognizes a 5'-NGG-3' protospacer-adjacent motif (PAM), directly fused to activation-induced cytidine deaminase AID* Δ . AID* Δ contained the P182X truncation together with three additional substitutions, K10E, T82I and E156G.

Multiple sgRNAs were designed against each of the A, B and C regions of the TOP1 coding sequence, from which one optimal sgRNA was selected for each region. The spacer-encoding DNA sequences used in this study were as follows: sgRNA1, 5'-GGCTAAGAAGAACCTTCCCG-3'; sgRNA2, 5'-GACTCCGAGCGATCTCGAGT-3'; and sgRNA3, 5'-GGACCCGGTCGAAATTAGGC-3'. These sequences represent the sgRNA spacers and do not include the PAM. Each target site was adjacent at its 3' end to a 5'-NGG-3' PAM. The three sgRNAs were individually cloned into the pLX-sgRNA vector (Addgene plasmid no. 50662), and the resulting constructs were verified by Sanger sequencing.

The sgRNA expression constructs were packaged into lentiviral particles and introduced into Raji B cell pools stably expressing TOP1 and dCas9-AID*Δ. Cells were maintained and serially passaged for 14 days in RPMI-1640 medium supplemented with 10% fetal bovine serum, allowing sgRNA-guided dCas9-AID*Δ to continuously target the corresponding coding regions and accumulate mutations.

Sequencing and mutation identification

After 14 days of mutagenesis and co-culture, genomic DNA was extracted from cells that survived the final round of selection. The complete insert encoding the designed protein was amplified using primers targeting conserved regions of the vector backbone. The specificity of the amplified product was verified by agarose gel electrophoresis, followed by gel extraction and purification. A targeted amplicon library was subsequently prepared and subjected to long-read sequencing.

The substitution frequency at each nucleotide position was calculated as the percentage of mutation-containing reads among all reads covering that position. A frequency of 0.1% was used as the operational threshold for identifying detectable mutations. Changes in variant frequencies before and after selection were used to evaluate the enrichment of specific mutations under functional selection pressure.

Generation of STAR-expressing macrophages

The STAR coding sequence was fused in frame to the intracellular domain of human FcRγ and cloned into the pCDH-EF1α-MCS-T2A-Puro lentiviral expression vector. The resulting lentiviral vector was co-transfected into HEK293T cells with the packaging plasmid psPAX2 and the envelope plasmid pMD2.G at a mass ratio of 4:3:1 to produce recombinant lentiviral particles.

THP-1 cells were transduced with concentrated lentiviral particles in the presence of 8 μg ml⁻¹ polybrene. At 72 h after transduction, the cells were selected with 2 μg ml⁻¹ puromycin for 7 days to establish a stable cell pool. Cell-surface expression of STAR was confirmed by flow cytometry using an anti-FLAG antibody.

SPR binding and competition assays

Surface plasmon resonance (SPR) experiments were performed using a Biacore instrument. For affinity measurements, purified STAR was immobilized on a CM5 sensor chip by amine coupling. Serial dilutions of the IL-6/sIL-6R α complex (0-2,000 pM) were injected over the chip surface at a flow rate of 30 $\mu\text{l min}^{-1}$.

For competition assays, recombinant gp130 was immobilized on the sensor chip. The IL-6/sIL-6R α complex (2 nM) was preincubated with increasing concentrations of STAR (0, 0.3, 0.6, 0.9, 1.2, 1.5, 2 and 3 nM) for 30 min at room temperature and subsequently injected over the gp130-immobilized surface. Response units (RU) were recorded, and the data were analyzed using the Biacore evaluation software. The apparent equilibrium dissociation constant (K_D) was calculated by fitting the equilibrium responses to a 1:1 binding model.

Internalization assay

The IL-6/sIL-6R α complex was biotinylated and subsequently labeled with Alexa Fluor 488-conjugated streptavidin (SA-AF488; Invitrogen, S11223). STAR-Trap and WT-M ϕ cells (2×10^5 cells per condition) were incubated with the AF488-labeled complex for 30 min at 4 °C to allow cell-surface binding. After washing, the cells were transferred to 37 °C and incubated for 0, 15, 30, 60 or 120 min to initiate internalization.

Internalization was terminated by placing the cells on ice and washing them twice with ice-cold FACS buffer consisting of PBS supplemented with 1% bovine serum albumin and 2 mM EDTA. Each sample was then divided into two equal aliquots. One aliquot was left unquenched to measure total cell-associated fluorescence. The other aliquot was incubated with an anti-Alexa Fluor 488 polyclonal antibody (Invitrogen, A-11094) at 10 $\mu\text{g ml}^{-1}$ for 30 min on ice to quench fluorescence from surface-bound AF488. Cells were washed and analyzed by flow cytometry.

Confocal fluorescence microscopy

STAR-Trap and control THP-1-derived macrophages were incubated with the fluorescently labeled IL-6/sIL-6R α complex. After incubation, cells were washed to remove unbound fluorescent material. Lysosomal compartments were labeled with Lyso-Tracker Red (Beyotime,

C1046), and nuclei were counterstained with DAPI staining solution (Beyotime, C1006). Images were acquired by confocal microscopy to examine the intracellular distribution of the fluorescent signal and its colocalization with Lyso-Tracker-positive compartments.

Analysis of macrophage polarization

STAR-Trap and WT-M ϕ cells were collected and incubated with a human Fc receptor-blocking reagent (BD Biosciences) for 15 min at 4 °C. For cell-surface staining, cells were incubated for 30 min at 4 °C in the dark with fluorophore-conjugated antibodies against CD86 (clone BU63; BD Biosciences, catalog no. 572835), HLA-DR (clone L243; BD Biosciences, catalog no. 568231) and CD206 (clone 15-2 [MR15-2]; BD Biosciences, catalog no. 572493).

For intracellular staining of Arg-1 (clone 7D1D10; BD Biosciences, catalog no. 572285) and IL-10 (BD Biosciences, catalog no. 567013), cells were fixed and permeabilized using the BD Perm/Wash buffer system (catalog no. 554723) according to the manufacturer's instructions. Intracellular ROS were detected using the CM-H2DCFDA probe (Beyotime, catalog no. S0035M). Stained cells were analyzed by flow cytometry, and the data were processed using FlowJo software. Isotype controls were used to establish gating boundaries.

Assessment of IL-6 trans-signaling and classical signaling

To evaluate the effect of STAR-Trap on IL-6 trans-signaling, STAR-Trap were incubated with the IL-6/sIL-6R α complex for 24 h. Cell-free supernatants were collected and transferred to cultures of human umbilical vein endothelial cells (HUVECs), which are gp130-positive endothelial responder cells with low or undetectable expression of membrane-bound IL-6R α . After 15 min of stimulation, STAT3 phosphorylation at Tyr705 was measured by flow cytometry following fixation and permeabilization. MCP-1 and ICAM-1 levels were measured by flow cytometry or ELISA.

To evaluate classical IL-6 signaling, STAR-Trap were incubated with free IL-6 alone. Cell-free supernatants were collected and transferred to U266 cells expressing both membrane-bound IL-6R α and gp130. Phosphorylated STAT3, MCP-1 and ICAM-1 were measured as described above. Tocilizumab was included as a positive control for inhibition of classical IL-6 signaling.

All experiments were performed in triplicate wells, and the data were analyzed using GraphPad Prism.

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